Stats 20 Midterm Exam

Version A

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Introduction to Statistical Programming with R
UCLA

Instructions: Please turn your phone into airplane mode (or turn it off) and place it in your bag. You will have 65 minutes to complete this exam.

Please write your responses for the **free response questions** directly on the blank pages and mark each page with your name and student ID number.

Mark your answers to all **multiple choice** questions on the scantron provided and make sure you have as well as your name and student ID number on the file. Choose the one best option offered and mark your choice clearly on the provided scantron; ambiguous responses will be given 0 points.

There are 5 free response questions and a series of multiple choice questions worth 2 point each for a **total of 100 points**.(This practice exam has a few more for extra practice)

You hereby understand and agree to abide by the instructions above and UCLA's policy on academic integrity. All work you submit is a product of your own honest effort.

All questions on this exam should be done using only functions or syntax discussed in the lecture notes, homework, or required reading. No credit will be given for use of outside functions. At the end of this exam you will find a list of allowed functions.

Academic Misconduct: Any potential violation of UCLA's policy on academic integrity will be reported to the Office of the Dean of Students. All work on this exam must be your own. You may be recorded before, during, and after the exam.

Note: Any and all course material may not be posted online or shared with anyone at any time without explicit written permission by the instructor (Joshua Sparks), even after the quarter is over. Failure to comply is a breach of academic integrity.

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Name (PRINT CLEARLY):

UID:

Section (Circle One): Lecture 1 Lecture 2

Signature:

(Turn over when exam starts)

Free Response Questions

Problem 1

(16 points total) A small weather station tracks air quality data for 7 consecutive days. The data is stored in the following vectors:

```
date <- c("2025/4/18", "2025/4/19", "2025/4/20", "2025/4/21", "2025/4/22",
          "2025/4/23", "2025/4/24")
pm25 <- c(NA, 35, 60, 40, 20, NA, 30) # PM2.5 concentration in µg/m³
wind_speed <- c(5, 3, 4, NA, 6, 2, NA) # Wind speed in km/h
alert_day <- c(TRUE, FALSE, TRUE, FALSE, FALSE, TRUE, FALSE) # Was a national weather alert issued?
```

Each index corresponds to a single day. So `date[1]`, `pm25[1]`, `wind_speed[1]`, and `alert_day[1]` describe Day 1.

(a) (3 points) Write code that identifies the dates with missing PM2.5 measurements. Store these dates in a vector called `missing_pm25_dates`.

(b) (3 points) Write R code to compute the mean PM2.5 level, ignoring missing values.

(c) (6 points) Are days with national weather alerts more likely to have missing PM2.5 measurements? Write R code to compute the proportions of missing PM2.5 measurements on alert and non-alert days and compare which proportion is higher.

(d) (4 points) Write code to compute the number of days with complete data (no `NA`s in either `pm25` or `wind_speed`)?

The famous (Fisher's or Anderson's) iris data set gives the measurements in centimeters of the sepal (green and leaflike leaves surrounding the colorful part of the flower) length and width and petal (colorful "leaves" of a flower) length and width in centimeters, respectively, for 50 flowers from each of 3 species of iris. The species are Iris setosa, Iris versicolor, and Iris virginica.

Problem 2

(14 points total) Consider the following vectors, which contain various measurements on these 150 iris flowers (only the first 8 elements of each vector are shown for brevity):

```
# show only first eight entries of each vector
sepal_length[1:8]
```

```
[1] 5.1 4.9 4.7 4.6 5.0 5.4 4.6 5.0
```

```
sepal_width[1:8]
```

```
[1] 3.5 3.0 3.2 3.1 3.6 3.9 3.4 3.4
```

```
petal_length[1:8]
```

```
[1] 1.4 1.4 1.3 1.5 1.4 1.7 1.4 1.5
```

```
petal_width[1:8]
```

```
[1] 0.2 0.2 0.2 0.2 0.2 0.4 0.3 0.2
```

```
# first 50 entries are setosa, entries 51:100 are versicolor, entries 101:150 are virginica
species[c(1, 50, 51, 100, 101, 150)]
```

```
[1] setosa     setosa     versicolor versicolor virginica  virginica
Levels: setosa versicolor virginica
```

```
length(species)
```

```
[1] 150
```

You can assume that all of the vectors have the same length and there are no missing values. The ith entry in each vector corresponds to the same flower, i.e., the first flower is of Species "Iris setosa", has a sepal length of 5.1cm, a sepal width of 3.5cm, a petal length of 1.4cm, and a petal width of 0.2cm.

(a) (6 points) Create two vectors called `setosa_sepal_length` and `versicolor_sepal_length` containing the sepal lengths of all iris setosa flowers and iris versicolor flowers in the data set respectively.

(b) (5 points) Compute the test statistic t for an unpaired samples t-test for two means in R to test whether the average sepal length differs between the two species iris setosa and iris versicolor? Hint: Here is the formula:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}},$$

where $\bar{x}_i$ is the sample mean of the ith sample and s_i is the sample standard deviation of the ith sample, $i = 1, 2$.

(c) (3 points) You compute your test statistic $t = -10.52$. Based on this value, does there appear to be evidence that a null of hypothesis of "no difference in average sepal lengths between the two species" is false? Explain your reasoning.

Problem 3

(12 points total) Counting missing values.

(a) (4 points) Write code that counts the number of `NA` values (do **NOT** count `NaN`s) in a vector `x` defined as

```
# Hint: there are three NAs in this vector
x <- c(NA, -1, 0, 1, NA, NaN, NA, Inf, NaN, NaN, 1, 10)
```

and stores it in a new object called `n_NAs`:

(b) (3 points) Assuming an existing object `n_NAs` in your working directory that contains the correct number of missing values in `x`, write a command in R that will count the number of values that are **NOT** `NA`s in the same vector `x`.

(c) (5 points) Assuming an existing object `n_NAs` in your working directory that contains the correct number of `NA`s in `x`, write code that computes the test statistic z_{obs} for a one-proportion test defined as

$$z_{obs} = \frac{p - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

testing the Null hypothesis that the **proportion** of `NA`s is equal to 0.5 ($H_0 : p = 0.5$) against the alternative hypothesis that it is not ($H_a : p \neq 0.5$).

Note: You do not actually have to compute the value of the test statistic, just write the R code that will compute it.

Hint: I recommend defining the variables p, p_0, and n as a first step. Then implement the formula for z_{obs} using these variables.

Problem 4

(14 points total) Consider the following vector of measurements, in millimeters:

```
x_mm<- c(74, 86, 91, 90, 67)
```

(a) **(3 Points)** Suppose I want to transform this vector from millimeters to meters (so, divide by 1000). Write the code for the new vector (x_m) using R's syntax for scientific notation

(b) **(3 Points)** In R, if we were print element x_m[2] out to twenty digits to the right of the decimal, would this value read 0.08600000000000000000 (or just 0.086)? Explain why or why not, directly mentioning how our personal number system differs from the number system R uses.

(c) **(4 Points)** Examine the R output below.

```
x_mm*(34/1000)/34==x_m
```

```
[1]  TRUE FALSE FALSE  TRUE  TRUE
```

Explain briefly why this output occurred. Use precise language used in class.

(d) **(4 Points)**

Suppose we wish to take the fifth element of `x_mm` and raise it to the 10000th power as such:

```
x_mm[5]^10000
```

What should the output be and explain your reasoning, including if it doesn't align "mathematically".

Problem 5

(24 points total) Your friend is about to hand in their homework where they implemented the following variance. It works, but it is a bit of a mess and does not make much sense as to what is happening.

```
numeric <- c(1,3,2,5,6)
MyValues=numeric
a=(length(MyValues)-1)
b=1/a; b2=mean(numeric)
c=(b2 - MyValues)
    cp2=c^2
cp2s=sum(cp2)
  Fin_res= cp2s*1 / a
# believe it or not, this actually works
Fin_res
```

```
[1] 4.3
```

(a) **(10 Points)** To begin, translate the steps made in the above code into a pseudo-code that explains the process fully.

(b) **(12 Points)** Please help them rewrite this code that provides clear names of the objects defined and includes commenting that guides the process. Fix at least 6 mistakes and comment them (there are way more than that) for full credit.

(c) **(2 Points)** Explain briefly (1-2 short sentences) why this two-pass formula provided is preferred over $s^2 = \frac{1}{n-1}(\sum_{i=1}^{n}(x_i^2) - n\bar{x}^2)$.

Multiple Choice Questions

Mark your answers to all multiple choice questions on the exam provided. Each multiple choice question is worth 1 point.

Note: While there are more than 20 questions here (adding up to more than 100 points), it is only to give you more practice. Overall, you should expect only 100 points on the exam.

Problem 1

What is the output of the following code:

```
x <- -4:4
x[x < 0 | x > 0]
```

A. `[1] -4 -3 -2 -1  1  2  3  4`

B. `[1] -4 -3 -2 -1  0  1  2  3  4`

C. `[1] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE`

D. `integer(0)`

E. An error message.

Problem 2

What is the output of the following code:

```
x <- -4:4
x[x < 0 & x > 0]
```

A. `[1] -4 -3 -2 -1  1  2  3  4`

B. `[1] -4 -3 -2 -1  0  1  2  3  4`

C. `[1] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE`

D. `integer(0)`

E. An error message.

Problem 3

What is the output of

```
x <- c(-1, 0, 1, 2, NA)
y <- c(-1, 0, NA, 2, 3)
x | y
```

A. `[1] TRUE`

B. `[1]  TRUE FALSE  TRUE  TRUE  TRUE`

C. `[1] TRUE TRUE TRUE TRUE TRUE`

D. `[1]  TRUE FALSE    NA  TRUE    NA`

E. `[1]  TRUE FALSE  TRUE  TRUE    NA`

Problem 4

What is the output of

```
x <- c(-1, 0, 1, 2, NA)
y <- c(-1, 0, NA, 2, 3)
x & y
```

A. `[1] TRUE`

B. `[1]  TRUE FALSE  TRUE  TRUE  TRUE`

C. `[1] TRUE TRUE TRUE TRUE TRUE`

D. `[1]  TRUE FALSE    NA  TRUE    NA`

E. `[1]  TRUE FALSE  TRUE  TRUE    NA`

Problem 5

What is the output of

```
x <- c(-1, 0, 1, 2, NA)
y <- c(-1, 0, NA, 2, 3)
x[!is.na(y) & y > 0]
```

A. `[1] 1 2`

B. `[1]  2 NA`

C. `[1] NA  2`

D. `[1] FALSE FALSE  TRUE  TRUE FALSE`

E. `[1] FALSE FALSE FALSE  TRUE  TRUE`

The following information is used in Problems Problem 6 and Problem 8.

Consider the following command:

```
x <- c(-3, 4 / 9, 2.5, 0, NA)
```

Problem 6

Which of the following commands will extract the number `0` from the vector `x`?

A. `x[4]`

B. `x(4)`

C. `x{4}`

D. `x[-4]`

E. `x(-4)`

Problem 7

Which of the following commands will extract everything but the number `-3` from the vector `x`?

A. `x[-1]`

B. `x[-c(-1)]`

C. `x[-3]`

D. `x[-c(-3)]`

E. `x[4:length(x)]`

Problem 8

What is the output of the following command

```
x <- c(-3, 4 / 9, 2.5, 0, -1, NA)
length(x)
```

A. `[1] 0`

B. `[1] 1`

C. `[1] 5`

D. `[1] 6`

E. `[1] 7`

Problem 9

What is the output of the following command

```
x <- c(-3, 1, 2.2, 0, -1)
x[x]
```

A. `[1] 1.0 0.0`

B. `[1] 1.0  0.0 -1.0`

C. `[1] -3.0  1.0`

D. `[1] -3.0  1.0  0.0 -1.0`

E. An error message.

Problem 10

What is the output of the command

```
c(FALSE, 3, 1, TRUE)
```

A. `[1] "FALSE" "3" "1" "TRUE"`

B. `[1] 0 3 1 1`

C. `[1] 1 3 1 0`

D. `[1] FALSE 3 1 TRUE`

E. `[1] FALSE TRUE TRUE TRUE`

Problem 11

What is the output of the command

```
sum(is.na(c(TRUE, FALSE, NA, NA, NaN)))
```

A. `[1] 1`

B. `[1] 2`

C. `[1] 3`

D. `[1] 4`

E. `[1] 5`

Problem 12

Which of the following would **NOT** be a valid name for an object in R?

A. `ILoveStats20`

B. `i.love.stats.20`

C. `1L0v3Stats20`

D. `i_love_stats_20`

E. `il0v3stats20`

Problem 13

What is the output of the following code:

```
obj <- c("this", "is", "content")
obj[length(obj) + 1] <- c("this is more content")
obj
```

A. `[1] "this" "is" "content"`

B. `[1] "this" "is" "content" "this is more content"`

C. `[1] "this" "is" "more" "content" "this" "is" "more" "content"`

D. `[1] "this" "is" "more" "content"`

E. An error message.

Problem 14

What is the output of the following code

```
x1 <- c(0, 1, 2)
x2 <- c(0, -1, 2)
x1 * x2
```

A. `[1]  -5`

B. `[1]  0 1 -0.5`

C. `[1]  0 -1 4`

D. `[1]  0 0 0 0 -1 -2 0 2 4`

E. `[1]  0 0 0 0 -1 2 0 -2 4`

Problem 15

What is the output of the following code

```
x1 <- c(0, 1, 2, 3)
x2 <- c(0, -1, 2)
x1 * x2
```

A. `[1]  0 -1  4` with a warning message

B. `[1]  0 -1  4` without a warning message

C. `[1]  0 -1  4  0` with a warning message

D. `[1]  0 -1  4  0` without a warning message

E. An error message

Problem 16

What is the output of the command

```
rep(c(3, 1, 4), times = 2)
```

A. `[1] 3 1 4 2`

B. `[1] 3 3 1 1 4 4`

C. `[1] 3 1 4 3 1 4`

D. `[1] 3 1 4 2 3 1 4 2`

E. An error message.

Problem 17

What is the output of the command

```
rep(c(3, 1, 4), each = 2)
```

A. `[1] 3 1 4 2`

B. `[1] 3 3 1 1 4 4`

C. `[1] 3 1 4 3 1 4`

D. `[1] 3 1 4 2 3 1 4 2`

E. An error message.

Problem 18

You want to randomly assign **exactly** half of your 50 subjects to the treatment group and the other half to the control group. How can you create a character vector that contains each subject's randomly assigned group?

A. `sample(x = c("treatment", "control"), size = 50, replace = FALSE)`

B. `sample(x = c("treatment", "control"), size = 50, replace = TRUE)`

C. `sample(x = rep(c("treatment", "control"), each = 25), size = 50, replace = FALSE)`

D. `sample(x = rep(c("treatment", "control"), each = 25), size = 50, replace = TRUE)`

E. `rep(c("treatment", "control"), each = 25)`

Problem 19

Which of the following will produce the same output as

```
-5 + 3:5 * 2
```

A. `((-5 + 3):5) * 2`

B. `(-5 + 3):(5 * 2)`

C. `(-5 + (3:5)) * 2`

D. `-5 + (3:(5 * 2))`

E. `-5 + ((3:5) * 2)`

Problem 20

Which of the following is meant by

```
3.2 * exp(4)
```

A. 3.2^4

B. $3.2 \times e^4$

C. $3.2 \times e^{-4}$

D. 3.2×2^4

E. 3.2×10^4

Problem 21

What is the output of

```
c(1, 3, 5) + c(1, 2)
```

A. `[1] 2 5` **without** a warning message

B. `[1] 2 5` with a warning message

C. `[1] 2 5 6` **without** a warning message

D. `[1] 2 5 6` with a warning message

E. None of the above: R will throw an error message

Problem 22

Which of the following does **NOT** produce the same output as

```
seq(3, 10, by = 2)
```

A. `c(3L, 5L, 7L, 9L)`

B. `(3:10)[c(TRUE, FALSE)]`

C. `seq_along((3:10)[c(TRUE, FALSE)])`

D. `(3:10)[(3:10) %% 2 == 1]`

Problem 23

Which code generates the following matrix?

```
     [,1] [,2] [,3] [,4]
[1,]    1    2    3    4
[2,]    1    2    3    4
[3,]    1    2    3    4
```

A. `matrix(rep(1:4, each = 3), nrow = 3, ncol = 4)`

B. `matrix(rep(1:4, each = 3), nrow = 4, ncol = 3)`

C. `matrix(rep(1:4, times = 3), nrow = 3, ncol = 4)`

D. `matrix(rep(1:4, times = 3), nrow = 4, ncol = 3)`

E. `matrix(rep(1:4, times = 4), nrow = 4, ncol = 3)`

Problem 24

Consider the following matrix `A`

```
A
```

```
     [,1] [,2] [,3] [,4]
[1,]    1    2    3    4
[2,]    5    6    7    8
[3,]    9   10   11   12
```

How would you extract the vector

```
[1] 11  7
```

from the matrix `A`?

A. `A[2:3, 2]`

B. `A[2:3, 3]`

C. `A[3:2, 3]`

D. `A[3, 2:3]`

E. `A[3, 3:2]`

Problem 25

What is the output of the following lines of code:

```
x <- 1:6
y <- x > 3
y & c(TRUE, FALSE, NA)
```

A. `[1] FALSE FALSE FALSE  TRUE FALSE FALSE`

B. `[1] FALSE FALSE FALSE  TRUE FALSE    NA`

C. `[1] FALSE FALSE FALSE  TRUE FALSE  TRUE`

D. `[1]  TRUE FALSE    NA  TRUE  TRUE  TRUE`

E. `[1]  TRUE FALSE FALSE  TRUE  TRUE  TRUE`

Congratulations, you have reached the end of the exam!